

Subject Description Form

Subject Code	EIE4428
Subject Title	Multimedia Communications
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	EIE3333 Data and Computer Communications
Objectives	To study the technical issues and system solutions for providing multimedia communications on the Internet.
Intended Subject Learning Outcomes	Upon completion of the subject, students will be able to: <u>Category A: Professional/academic knowledge and skills</u> 1. Understand the current state-of-the-art developments in Internet technologies for multimedia communications. 2. Appreciate the principles used in designing multimedia protocols, and so understand why standard protocols are designed the way that they are. 3. Understand the system design principles of multimedia communications systems. 4. Solve problems and design simple networked multimedia systems. <u>Category B: Attributes for all-roundedness</u> 5. Think critically and learn independently.
Subject Synopsis/ Indicative Syllabus	Syllabus: 1. <u>Terminal/Codec Support for Multimedia Communications</u> Scalable Coding: SNR Scalability, Spatial Scalability, Temporal Scalability and Fine Granularity Scalability (FGS) Error Control: Error Propagation, Error Resilience Coding Techniques Rate Control: Concepts for Rate Control, MPEG TM5 Rate Control Algorithms 2. <u>Transport Layer Support for Multimedia Communications</u> TCP congestion control, TCP Delay Analysis, TCP Throughput Analysis, Bandwidth Allocation. Media transport protocols: Real Time Protocol (RTP) and Real Time Control Protocol (RTCP); Signalling Protocols: Real-Time Streaming Protocol (RTSP) 3. <u>Quality of Services (QoS)</u> Integrated services (intserv): Architecture and Service Model, Resource Reservation Protocol (RSVP), Packet Scheduling Disciplines in the Internet Differentiated Services (diffserv): Framework and Concept, Assured and Expedited Services, Packet Classification, Routers Internals and Packet Dropping Techniques 4. <u>Multimedia Streaming Systems</u> Streaming architecture: Real-time Streaming and On-demand Streaming, Content Delivery Network (CDN), Data Sharing Techniques, Support of Interactive

	Operations, Peer-to-Peer (P2P) video streaming techniques, Case Studies on Video on Demand and IPTV						
	Laboratory Experiments/Mini-projects: 1. Multimedia networking 2. Multimedia streaming						
Teaching/ Learning Methodology	Teaching and Learning Method	Intended Subject Learning Outcome	Remarks				
	Lectures	1, 2, 3	fundamental principles and key concepts of the subject are delivered to students				
	Tutorials	1, 2, 3, 4, 5	supplementary to lectures and are conducted with smaller class size; students will be able to clarify concepts and to have a deeper understanding of the lecture material; problems and application examples are given and discussed				
	Laboratory sessions/Mini-projects	4, 5	students will make use of network simulators to simulate various types of communication networks and evaluate their performance, or students will develop a simple multimedia streaming system by integrating different components together using some existing tools.				
Assessment Methods in Alignment with Intended Subject Learning Outcomes	Specific Assessment Methods/Tasks	% Weighting	Intended Subject Learning Outcomes to be Assessed (Please tick as appropriate)				
			1	2	3	4	5
	1. Continuous Assessment (total 50%)						
	• Assignments	8%	✓	✓	✓		✓
	• Mid-Term Test	13%	✓	✓	✓	✓	✓
	• End-of-Term Test	13%	✓	✓	✓	✓	✓
	• Mini-Project	16%				✓	✓
	2. Examination	50%	✓	✓	✓	✓	✓
	Total	100%					

	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:	
	Specific Assessment Methods/Tasks	Remark
	Assignments, tests and examination	end-of chapter type problems used to evaluate students' ability in applying concepts and skills learnt in the classroom; students need to think critically and creatively in order to come with an alternate solution for an existing problem
	Laboratory sessions / mini-projects	each group of students are required to produce a written report; accuracy and the presentation of the report will be assessed.
Student Study Effort Expected	Class contact (time-tabled):	
	• Lecture	24 Hours
	• Tutorial/Laboratory/Practice Classes	15 Hours
	Other student study effort:	
	• Lecture: preview/review of notes; homework/assignment; preparation for test/quizzes/examination	36 Hours
	• Tutorial/Laboratory/Practice Classes: preview of materials, revision and/or reports writing	30 Hours
	Total student study effort:	105 Hours
Reading List and References	Reference Books: 1. J.K. Kurose, <i>Computer Networking: A Top-down Approach Featuring the Internet</i>, 6th ed., Pearson, 2012. 2. Ze-Nian Li and Mark S. Drew and J. Liu, <i>Fundamentals of Multimedia</i>, Springer, 2nd Edition, 2014. 3. K.R. Rao, Z.S. Bojkovic and D.A. Milovanovic, <i>Multimedia Communication Systems: Techniques, Standards, and Networks</i>, Prentice-Hall PTR, 2002.	

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